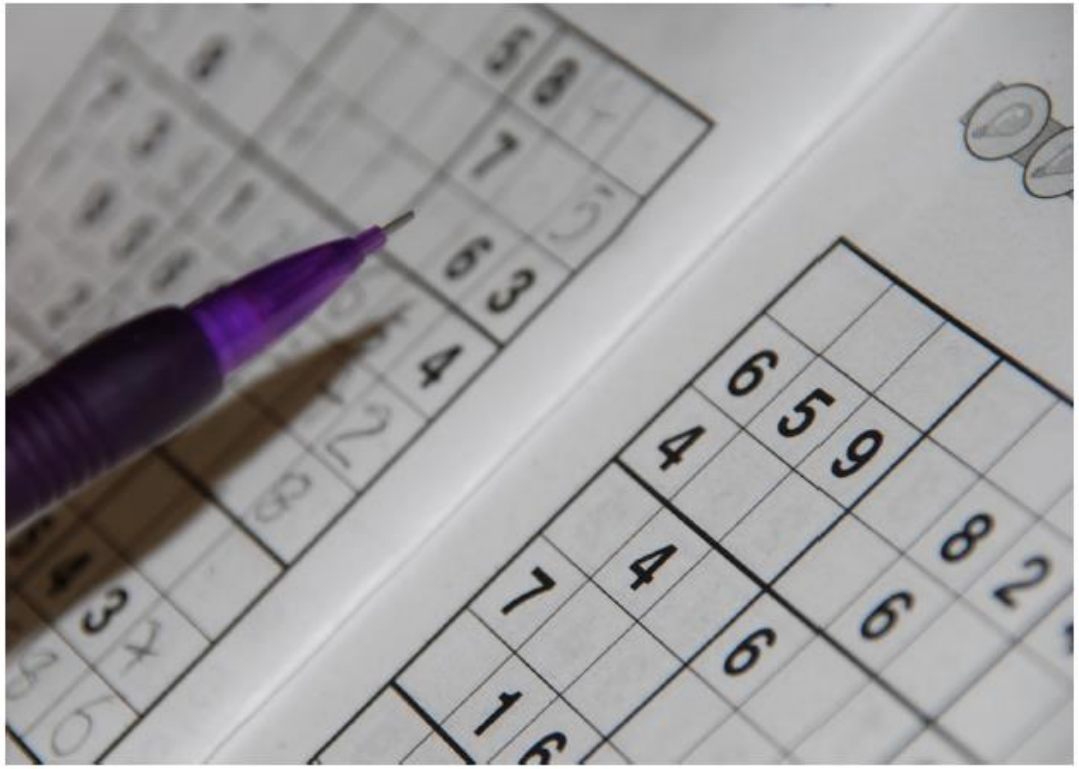
CSE358

ARTIFICIAL INTELLIGENCE

06. CONSTRAINT SATISFACTION PROBLEMS (CSP)

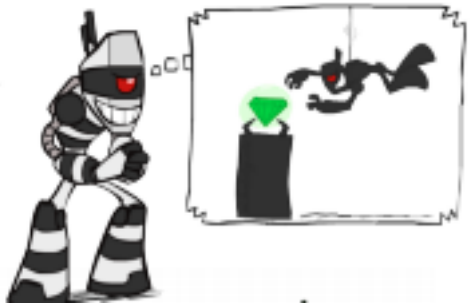
Outline

- Constraint Satisfaction Problems (CSPs)
- Crypt-arithmetic Puzzle
- Backtracking
- Constraint Propagation
- Problem Structure



Recall

Planning: sequences of actions



- **Search Problems**
  - Find the **sequence of actions** that leads to the goal
  - Sequence of actions means a **path** in the search space
  - The **path to the goal** is the important thing
  - Paths come with different costs and depths
  - We use "rules of thumb" aka **heuristics** to guide the search efficiently
- **Constraint Satisfaction Problems (CSP)**
  - A search problem too! But, a special subset of search problems
  - We care about the **goal itself**, ***not the path***
  - CSPs are specialized for identification problems

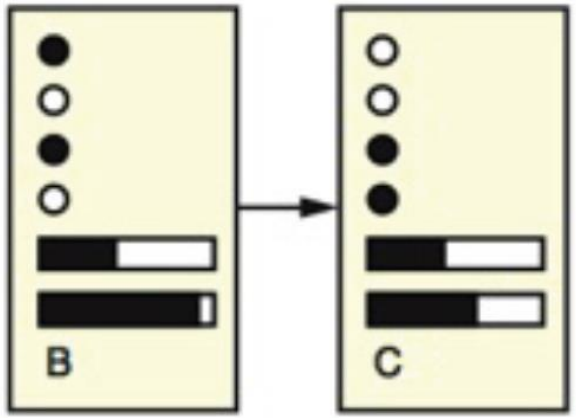
Identification: assignments to variables

Search vs. CSPs

Search Problems

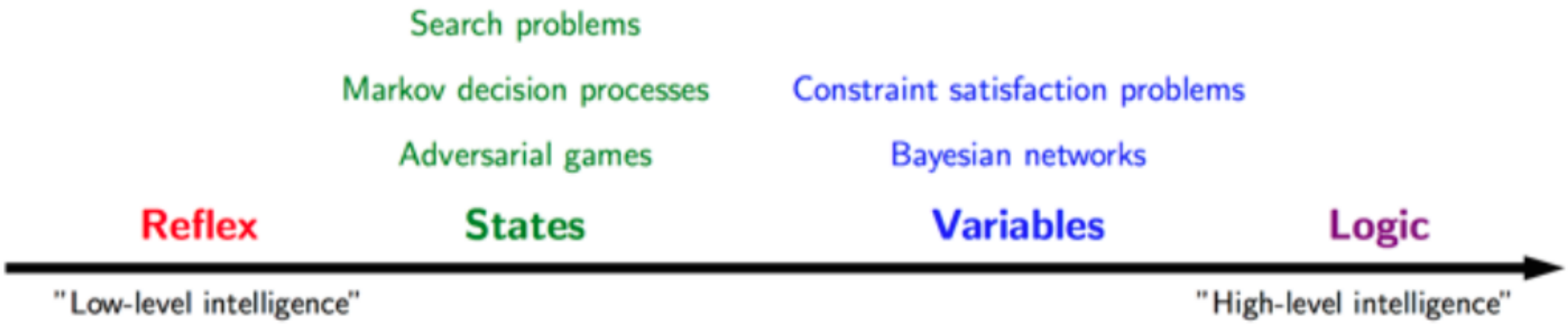
- A ***state*** is a **black box**, implemented as some data structure that supports successor function, heuristic function and goal test
- Recall **atomic representation**
- A **goal test** is any function over the states

CSP Problems



- A ***state***: defined by <u>variables</u> X_i with values from domain D_i → <u>attribute-value pairs</u>
- Recall **factored representation**
- A **goal test** is ***a set of constraints*** specifying ***allowable combinations*** of values for subsets of variables

Recall: The Intelligence Level Graph



Credit: Courtesy Percy Liang

CSPs Definition

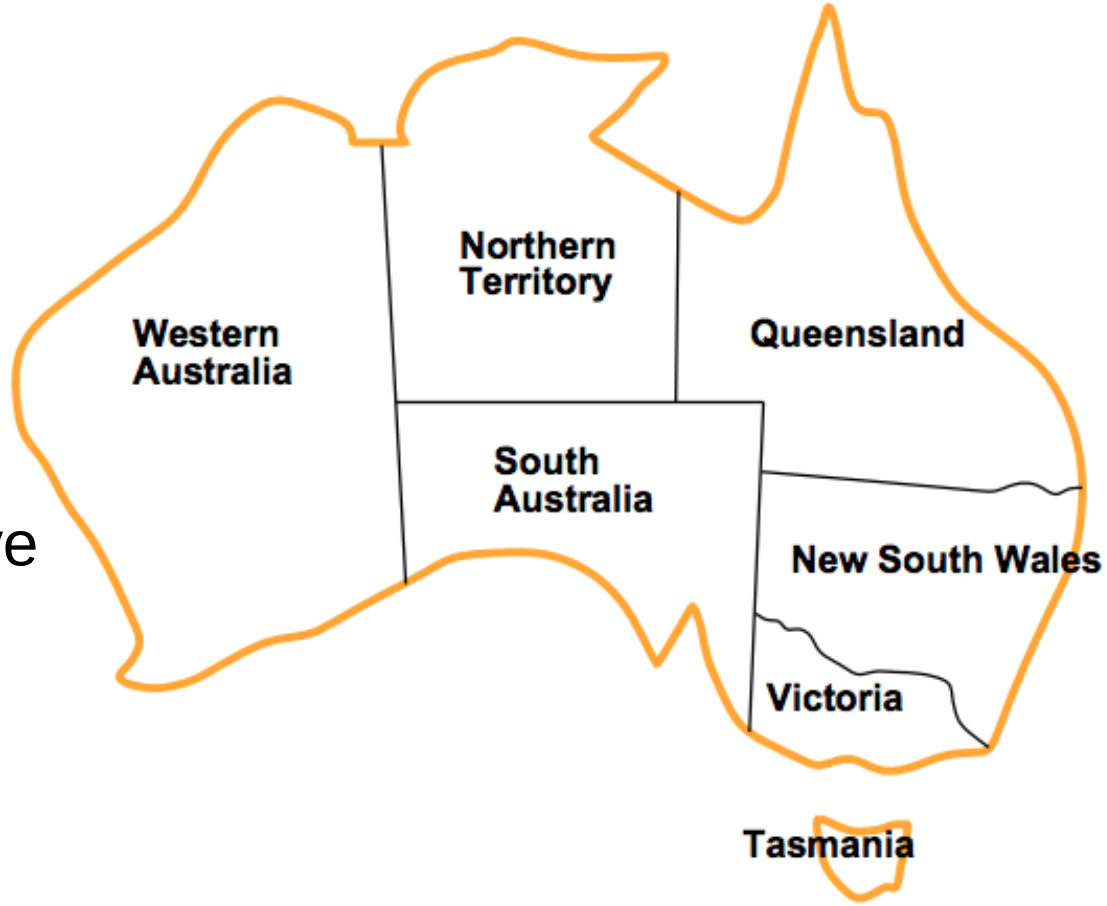
- A constraint satisfaction problem (CSP) consists of **three elements**:
  - A set of **variables**, $X = \{X_1, X_2, \ldots, X_n\}$
  - A set of **domains** (possible values) for each variable: $D = \{D_1, D_2, \ldots, D_n\}$
  - A set of **constraints** *C* that specify <u>allowable combinations</u> of values
- <u>Solving the CSP:</u> **finding variable assignment(s)** that **satisfy all constraints**
- <u>Concepts:</u> problem formalization, backtracking search, arc consistency, etc.
- We call "**a solution**" → a **consistent assignment**
  - Why consistent? – find the assignment that does not violate the constraints

Example: Map Coloring

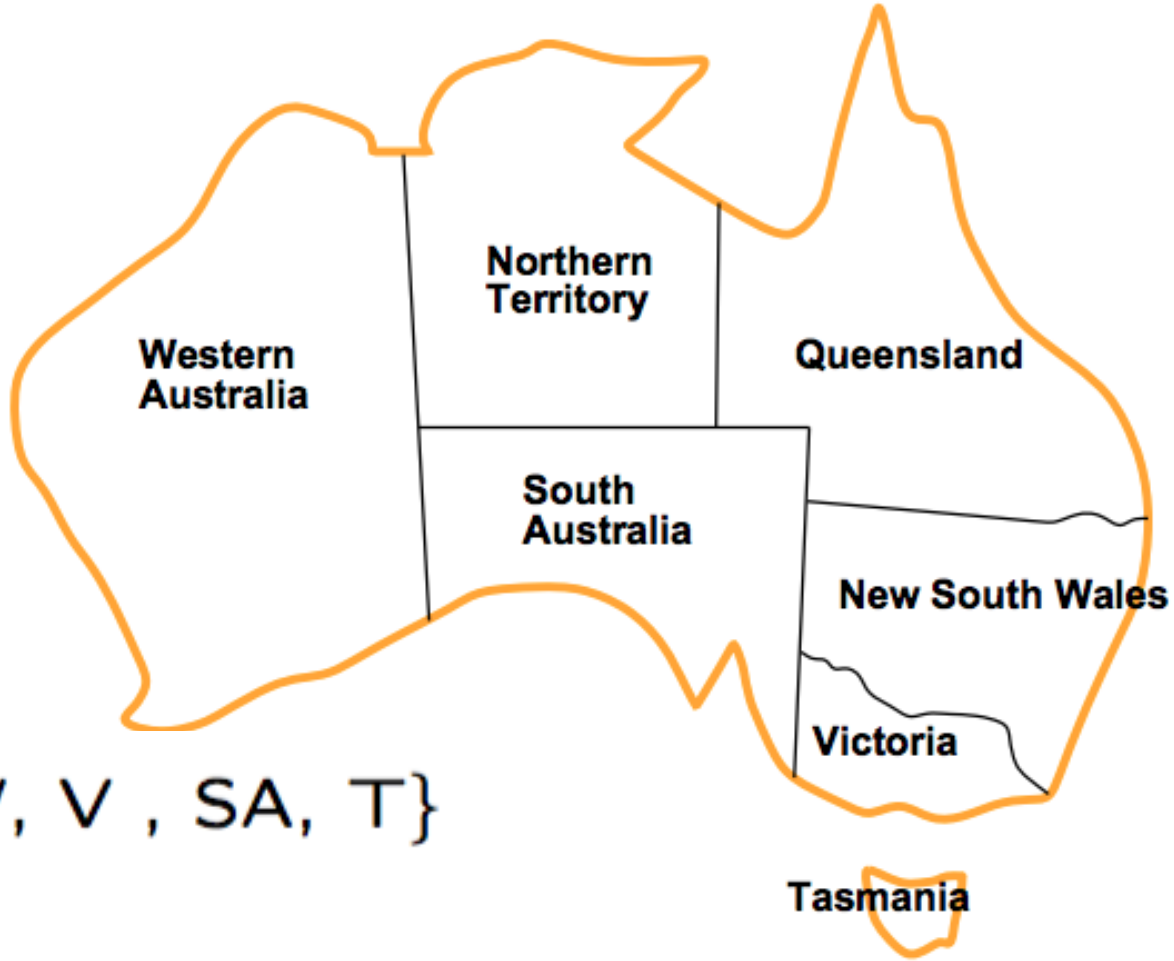
Question: How can we color each of the 7 provinces {**red**, **green**, **blue**} so that no two neighboring provinces have the same color?

Task: Color each region {**red**, **green**, **blue**}

Constraint: No two adjacent regions have the same color



Example: Map Coloring



Variables: X ={WA, NT, Q, NSW, V , SA, T}

Domains: D_i = {red, green, blue}

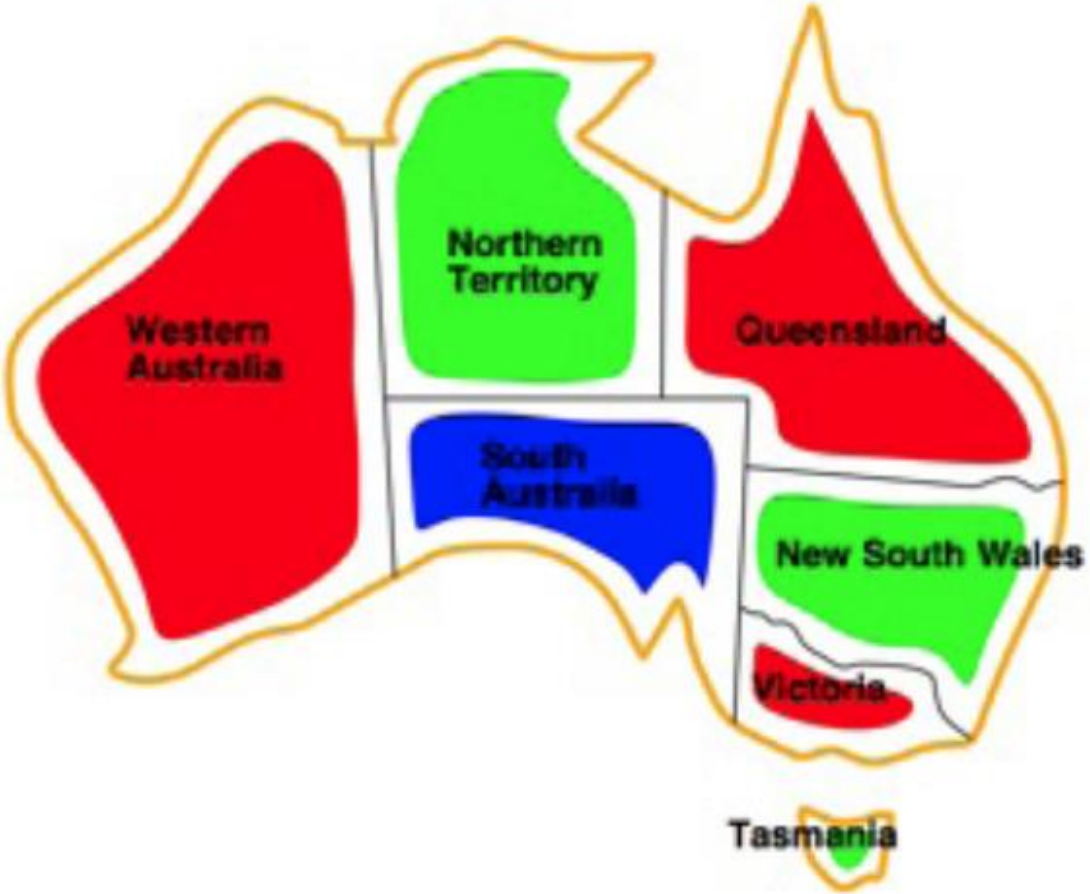
Constraints: adjacent regions must have different colors;

e.g., WA $\neq$ NT or (W A, N T) $\in$ {(red, green), (red, blue), etc..}

$$C = \{WA \neq SA, WA \neq NT, NT \neq SA, NT \neq Q, SA \neq Q, SA \neq NSW, SA \neq V, Q \neq NSW, NSW \neq V\}$$

Example: Map Coloring

One Possible Solution



Example:

{WA=red, NT=green, Q=red, NSW=green,V=red, SA=blue,T=green}

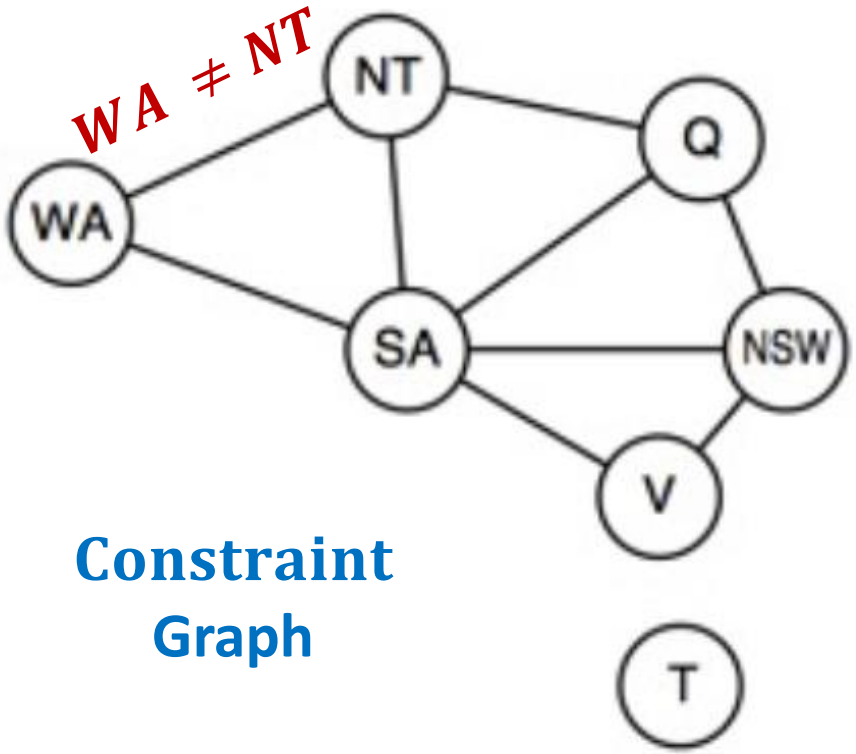
Real-World CSPs

- Assignment problems (e.g., who teaches what class?)
- Timetabling problems (e.g., which class is offered when and where?)
- Hardware configuration
- Spreadsheets
- Transportation scheduling
- Factory scheduling
- Floor planning

Notice that many real-world problems involve real-valued variables, rather than discrete variables such as the color problem in the map coloring

Constraint Graph

$$C = \{WA \neq SA, WA \neq NT, NT \neq SA, NT \neq Q, SA \neq Q, SA \neq NSW, SA \neq V, Q \neq NSW, NSW \neq V\}$$



Constraint Graph

- **Binary CSP:** each constraint relates at most two variables
- **Binary Constraint Graph:** nodes are variables, arcs (edges) show constraints
- **CSP algorithms:** use the graph structure to speed up search
  - E.g., Tasmania is an independent sub-problem!

Varieties of Variables

- **Discrete Variables**
  - **Finite Domains**
    - Assume $\boldsymbol{n}$ variables, $\boldsymbol{d}$ values, then the number of complete assignments is $\boldsymbol{O}(\boldsymbol{d}^{\boldsymbol{n}})$
    - E.g., map coloring, 8-queens problem
  - **Infinite Domains** (integers, strings, etc.)
    - Cannot enumerate all combinations of values
    - Need to use a constraint language
    - E.g., job scheduling (variables are start/end days for each job) - $T_1 + d \leq T_2$
- **Continuous Variables**
  - Common in operational research
  - Linear programming problems with linear or non-linear equalities/inequalities

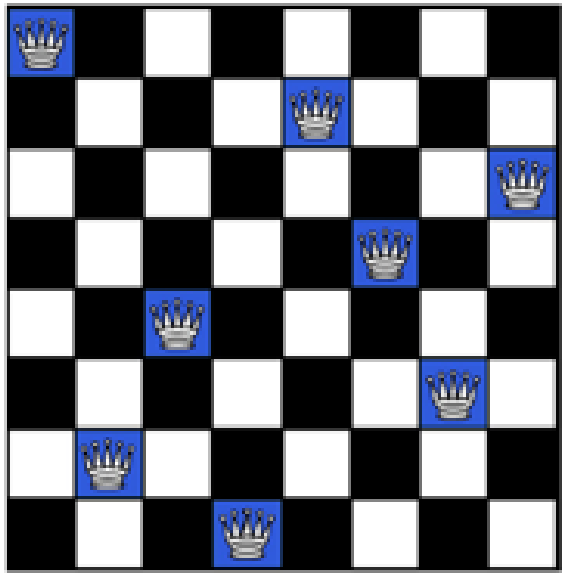
Varieties of Constraints

Constraints may have different scope:

- **Unary constraints:** constraint on a single variable, e.g., $\boldsymbol{SA} \neq \boldsymbol{green}$
- **Binary constraints:** constraints between pairs of variables, e.g., $\boldsymbol{SA} \neq \boldsymbol{WA}$
- **Global constraints:** involve 3 or more variables, e.g., ***Alldiff*** constraint specifies that all variables must have different values (e.g., of this kind of constraint involves, *crypt-arithmetic puzzles*, *Sudoku*)
- **Preferences (soft constraints):**
  - *Example:* red is better than green
  - Each variable assignment is associated with a **cost**
  - *Goal*: Find the assignment with the lowest overall cost that is consistent with all constraints
  - Involves problems such as "*constrained optimization problems*"

Example: 8-Queen Problem

8-Queen: Place 8 queens on an 8x8 chess board so no queen can attack another one

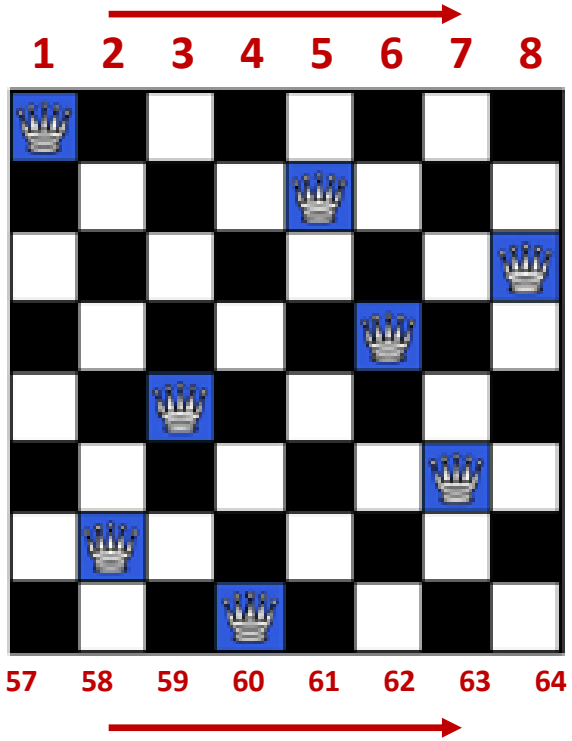


- Can treat this as a CSP
  - What are the variables, domains, and constraints?

Example: 8-Queen CSP

8-Queen: Place 8 queens on an 8x8 chess board

so, no queen can attack another one



- **Problem Formulation 1:**
  - **Variables:** 1 variable per queen

$$X = \{Q_1, Q_2, Q_3, Q_4, Q_5, Q_6, Q_7, Q_8\}$$

  - **Domains:** Each variable could have a value between 1 and 64

$$1 \leq Q_n \leq 64$$

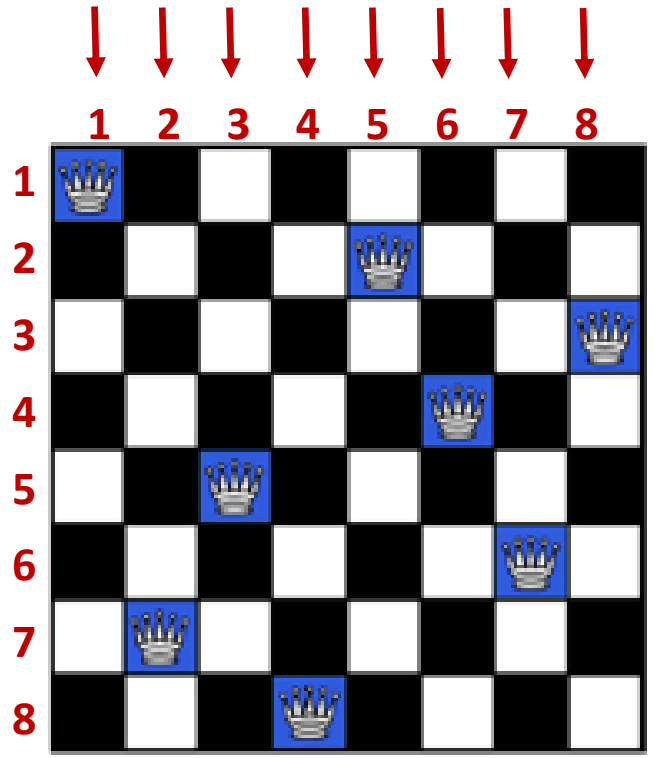
  - **Solution:**

$$Q_1 = 1, Q_2 = 13, Q_3 = 24, \dots, Q_8 = 60$$

  - **Constraints?**

Example: 8-Queen CSP

8-Queen: Place 8 queens on an 8x8 chess board so, no queen can attack another one



- **Problem Formulation 2:**
  - **Variables:** 1 variable per queen

$$X = \{Q_1, Q_2, Q_3, Q_4, Q_5, Q_6, Q_7, Q_8\}$$

  - **Domains:** Each variable could have a value between 1 and 8 (Row for the queen in the n-th column)

$$1 \leq Q_n \leq 8$$

  - **Solution:**

$$Q_1 = 1, Q_2 = 7, Q_3 = 5, \dots, Q_8 = 3$$

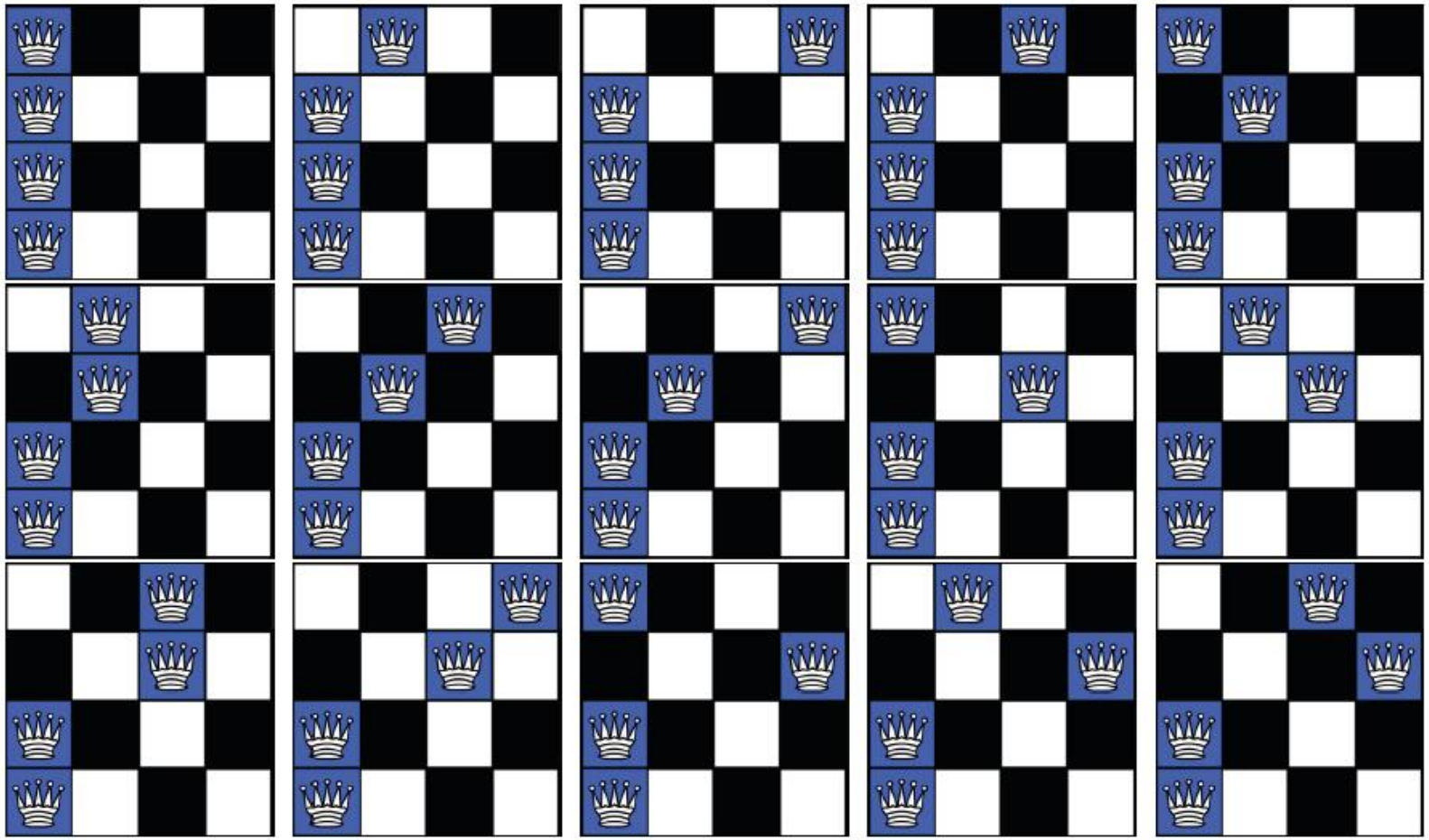
  - **Constraints:**

$$Q_1 \neq Q_2, Q_1 \neq Q_3, Q_1 \neq Q_4, \dots, Q_7 \neq Q_8$$

diagonals: $\forall (i, j) \quad |i - j| \neq |Q_i - Q_j|$

Brute Force?

Should we simply generate and test all configurations?



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- **4-Queen Problem**

  $4x4x4x4 = 4^4 = 256$

Is this brute force search scalable?

- **8-Queen Problem**

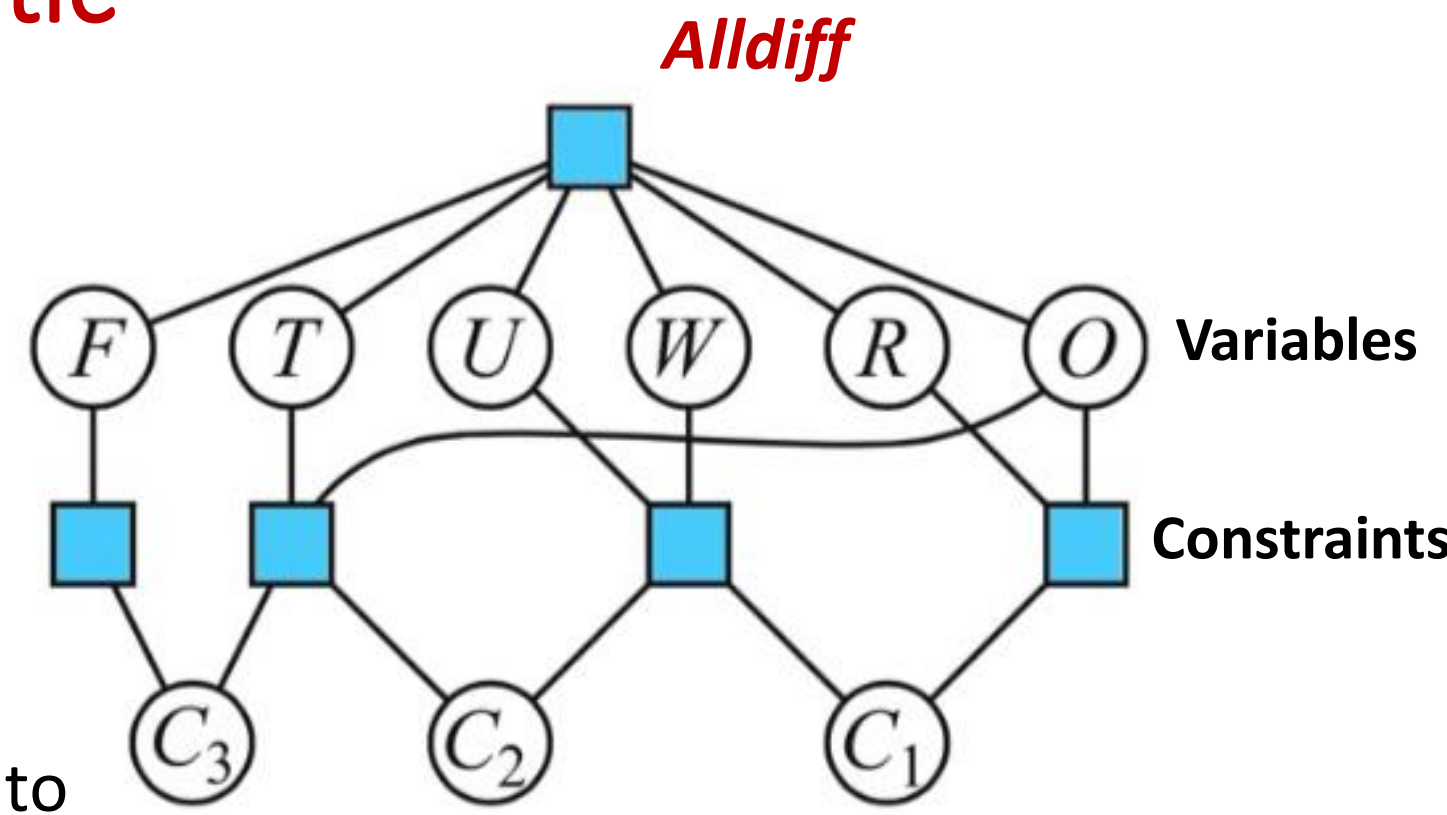
  $8x8x8x8x8x8x8x8 = 8^8$
  $= 16.7\ million$

- **16-Queen Problem**

  $16^{16} \approx 10^{20}$

Example: Crypt-arithmetic

$$\begin{array}{cccc} & C_3 & C_2 & C_1 \\ & T & W & O \\ + & T & W & O \\ \hline F & O & U & R \end{array}$$



Hypergraph

Crypt-arithmetic Puzzle: Problem is to find a substitution of letters with digits

- There's ten digits, such that we have the arithmetic sum here correct
- We want to find substitutions for T, W, O, F, U, and R

Example: Crypt-arithmetic

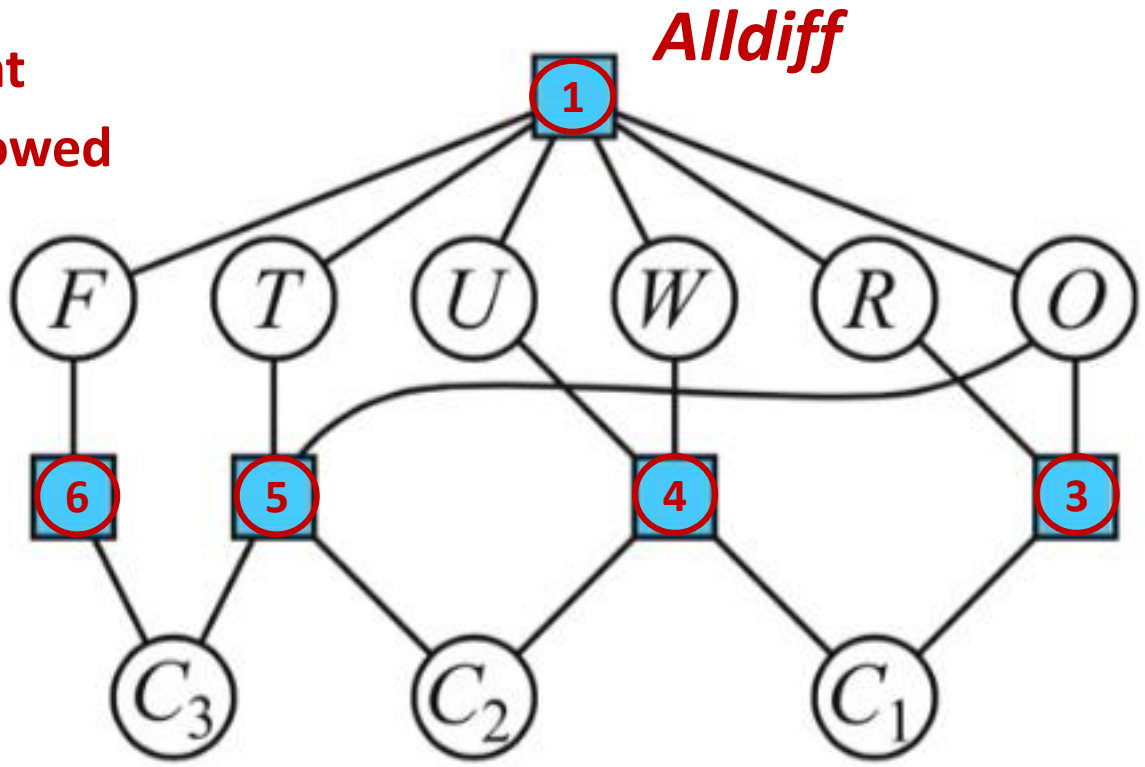
$$\begin{array}{rcccc} & c_3 & c_2 & c_1 & \\ & T & W & O & 734 \\ + & T & W & O & 734 \\ \hline F & O & U & R & 1468 \end{array}$$

Variables: $X = \{F, T, U, W, R, O, C_1, C_2, C_3\}$

Domain: $D = \{0, 1, 2, \cdots, 9\}$

Constraints:

1. $\text{Alldiff}(F, T, U, W, R, O)$ **n-ary constraint**
2. $T \neq 0, \ F \neq 0$ **leading zeros are not allowed**
3. $O + O = R + 10 * C_1$
4. $C_1 + W + W = U + 10 * C_2$
5. $C_2 + T + T = O + 10 * C_3$
6. $C_3 = F$



Solving CSPs

- **State-space search algorithms: search!**
- **CSP Algorithms:** Algorithm can do two things:
  - **Search:** choose a new variable assignment from many possibilities (e.g., backtracking search)
  - **Inference: constraint propagation**, *use the constraints to spread the word*: reduce the number of values for a variable which will reduce the legal values of other variables etc.
- Inference can be used
  - As a preprocessing step - constraint propagation sometimes can solve the problem entirely without search
  - But constraint propagation can also be intertwined with search in more difficult problems

Solving CSPs

- **BFS:** Develop the complete tree
- **DFS:** Fine, but time consuming
- **BTS: Backtracking search** is the basic uninformed search for CSPs. It's a **DFS** with these two improvements
  1. ***Assign one variable at a time:***
     - Variable assignments are commutative, so fix ordering
     - E.g., (WA=red then NT=green) is same as (NT=green then WA=red)
     - Only need to consider assignments to a single variable at each step
  2. ***Check constraints on the go:***
     - consider values that do not conflict with previous assignments

DFS with these two improvements is called BTS

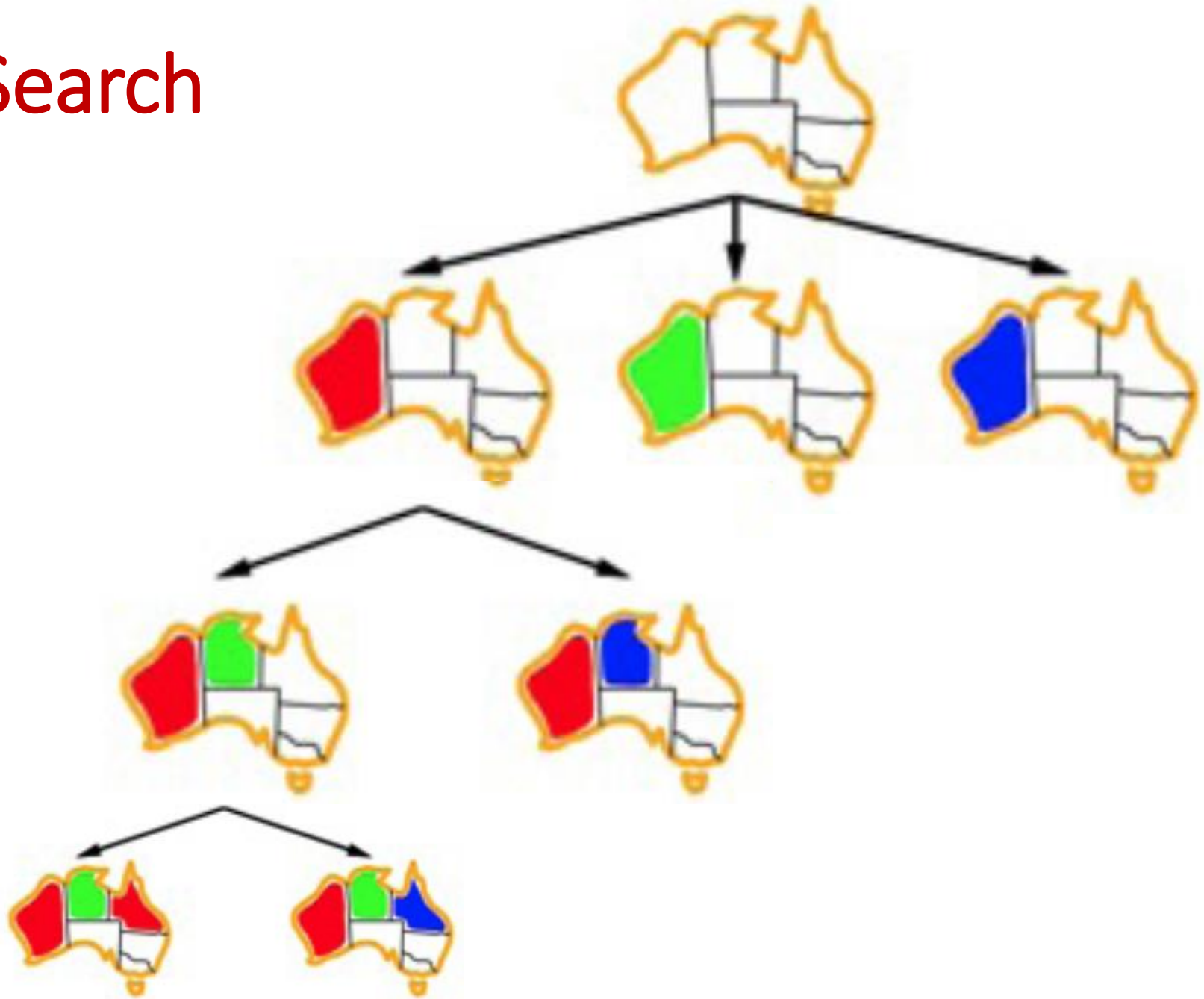
Typically, CSPs are a <u>DFS with some backtracking search</u>, which means that we're going to step back whenever we make an assignment that violates the previous assignments

Solving CSPs

Standard search formulation of CSPs

- **Initial state:** empty assignment {} - we did not assign any value to variables
- **States:** defined by the values assigned so far (partial assignments)
  - E.g., providing some variables with digits in the crypt-arithmetic puzzle
- **Successor function:** assign a value to an unassigned variable *(var=value)* that does not conflict with current assignment
  - Fails if no legal assignments
- **Goal test:** the current assignment is complete and satisfies all constraints – complete and consistent

Backtracking Search



Improving BTS

Heuristics are back!

1. **Which variable should be assigned next?** *(ordering)*
2. **In what order should its values be tried?** *(ordering)*
3. **Can we detect inevitable failure early?** *(filtering)*

Use General Purpose Heuristics, which give huge gains in speed

Minimum Remaining Values

1. **Which variable should be assigned next?**

- **MRV:** Choose the variable with the **fewest legal values in its domain**
- This is the variable that is most likely to cause failure. Fail first!

Pick the hardest!

Least Constraining Values

2. **In what order should its values be tried?**

- **LCV:** Given a variable, **choose the least constraining value:**
  - The one that rules out the <u>fewest values in the remaining variables</u>

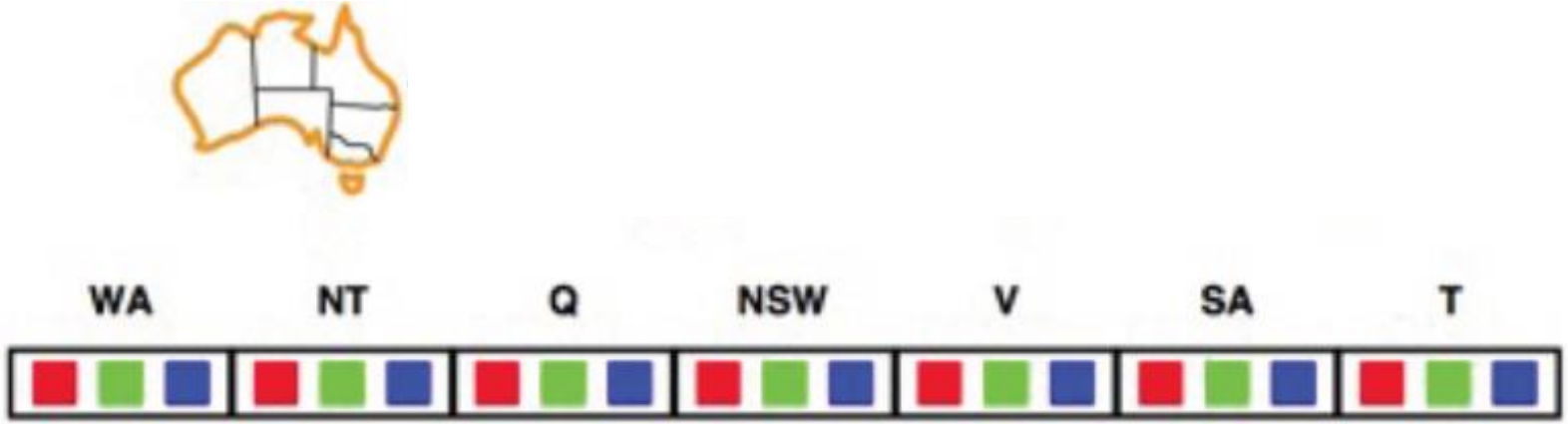
Allows 1 value for SA

Allows 0 values for SA

Pick the ones that are likely to work!

Forward Checking

3. **Can we detect inevitable/unavoidable failure early?**
   - **FC:** Keep track of <u>remaining legal values for unassigned</u> variables (using immediate constraints)
   - Terminate when any variable has no legal values



Helps to detect failure early

Forward Checking

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Helps to detect failure early

Backtracking Search

Input is **CSP formalization**, which means the domain, the variables, and the constraints

```
function BACKTRACKING_SEARCH(csp) returns a solution, or failure
    return BACKTRACK({}, csp)

function BACKTRACK(assignment, csp) returns a solution, or failure
    if assignment is complete then return assignment
    var = SELECT_UNASSIGNED-VARIABLE(csp)
    for each value in ORDER_DOMAIN_VALUES (var, assignment, csp)
        if value is consistent with assignment then
            add {var = value} to assignment
            result = BACKTRACK(assignment, csp)
            if result ≠ failure then return result
        remove {var = value} from assignment
    return failure
```

Pick a variable – one at a time

Backtracking = DFS + variable-ordering + fail-on-violation

Solving CSPs: Sudoku

All 3x3 boxes, rows, columns, **must contain all digits 1…9**

8		9	5		1	7	3	6
2		7		6	3			
1	6							
				9		4		7
	9		3		7		2	
7		6		8				
							6	3
			9	3		5		2
5	3	2	6		4	8		9

← **units**

Solving CSPs: Sudoku Formulization

All 3x3 boxes, rows, columns, **must contain all digits 1…9**

	1	2	3	4	5	6	7	8	9
A	8		9	5		1	7	3	6
B	2		7		6	3			
C	1	6							
D					9		4		7
E		9		3		7		2	
F	7		6		8				
G								6	3
H				9	3		5		2
I	5	3	2	6		4	8		9

Variables:

$$V = \{A_1, \dots, A_9, B_1, \dots, B_9, \dots, I_1, \dots, I_9\}, |V| = 81$$

Domain:

$D = \{1, 2, \dots, 9\}$, the filled squares have a single value

Constraints: 27 constraints

- $\text{Alldiff}(A_1, A_2, A_3, A_4, A_5, A_6, A_7, A_8, A_9)$ → ROWS
  $\dots$ (8 more)
- $\text{Alldiff}(A_1, B_1, C_1, D_1, E_1, F_1, G_1, H_1, I_1)$ → COLUMNS
  $\dots$ (8 more)
- $\text{Alldiff}(A_1, A_2, A_3, B_1, B_2, B_3, C_1, C_2, C_3)$ → 3x3 BOXES
  $\dots$ (8 more)

Solving CSPs: Sudoku

All 3x3 boxes, rows, columns, **must contain all digits 1…9**

8	~~2, 4~~	9	5	~~2, 4~~	1	7	3	6
2	~~3, 5~~	7		6	3			
1	6	~~3, 5~~						
				9		4		7
	9		3		7		2	
7		6		8				
							6	3
			9	3		5		2
5	3	2	6		4	8		9

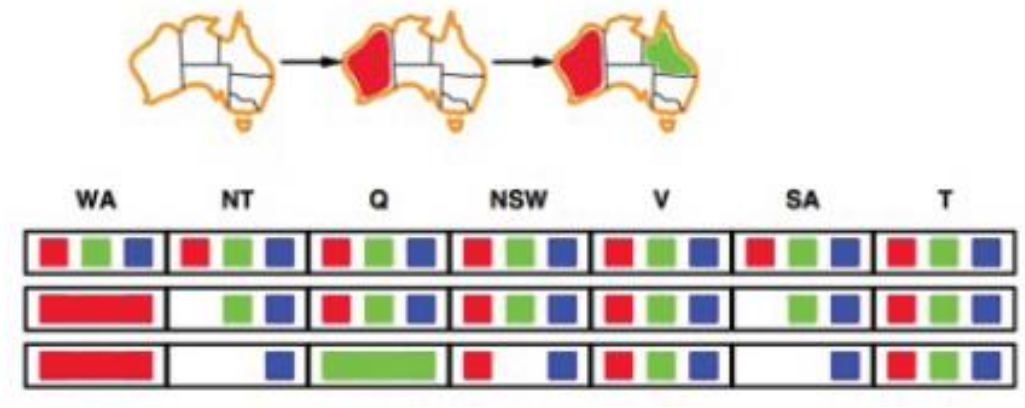
Constraint Propagation

8	4	9	5	2	1	7	3	6
2	5	7	8	6	3	9	1	4
1	6	3	7	4	9	2	5	8
3	2	5	1	9	6	4	8	7
4	9	8	3	5	7	6	2	1
7	1	6	4	8	2	3	9	5
9	8	4	2	7	5	1	6	3
6	7	1	9	3	8	5	4	2
5	3	2	6	1	4	8	7	9

- **Naked doubles (triples):** find two (three) cells in a 3x3 grid that have only the same candidates left, eliminate these two (three) values from all possible assignments in that box
- Locked pair, Locked triples, etc.

Constraint Propagation

- Forward checking propagates information from **assigned to adjacent unassigned variables, but doesn't detect more distant failures**
- Observe:



  - **Forward checking** does not check interaction between unassigned variables!
  - Here **SA** and **NT**! (They both must be blue, but they cannot both be blue!)
  - Why didn't we detect this yet?
- Forward checking improves backtracking search but does not look very far in the future, hence does not detect all failures, and sometimes can make mistakes
  - There is no propagation between unassigned variables
- We use **constraint propagation**, reasoning from constraint to constraint
  - *Constraint propagation* repeatedly enforces constraints (locally)
  - E.g., arc consistency test

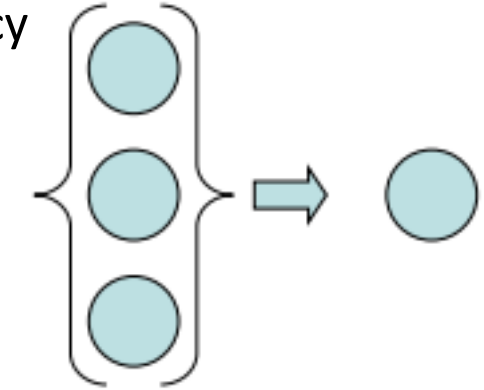
Types of Consistency

Constraint propagation aims to make the graph, or the CSP problem, consistent.
But there are different kinds of consistency – different levels

- **Node-consistency** (or **1-Consistency**) - (***unary constraints***): Each single node's domain has a value which meets that node's unary constraints
  - E.g, if we want Tasmania to be absolutely not green, add the unary constraint $T \neq green$, and use that constraint to make the node Tasmania consistent
  - If each single node in the CSP graph is node-consistent, which means cannot take any forbidden value → Node-consistency in CSP

- **Arc-consistency** (or **2-Consistency**) - (***binary constraints***): is the consistency between edges. For each pair of nodes, any consistent assignment to one can be extended to the other
  - To make a graph arc-consistent, check all possible edges for their arc-consistency status, if they are all arc-consistent, → call the problem arc-consistent

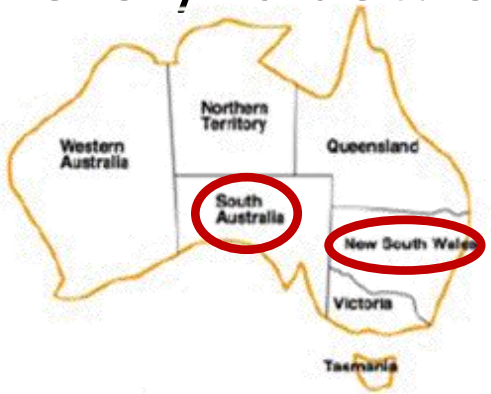
- **Path-consistency** (or **n-Consistency**) - (***n-ary constraints***): generalizes arc-consistency from binary to multiple constraints. For each n nodes, any consistent assignment to k-1 can be extended to the kth node
- Higher ***n*** more expensive to compute
- ***Note:*** It is always possible to transform all n-ary constraints into binary constraints
  - Often, CSPs solvers are designed to work with binary constraints



Arc Consistency

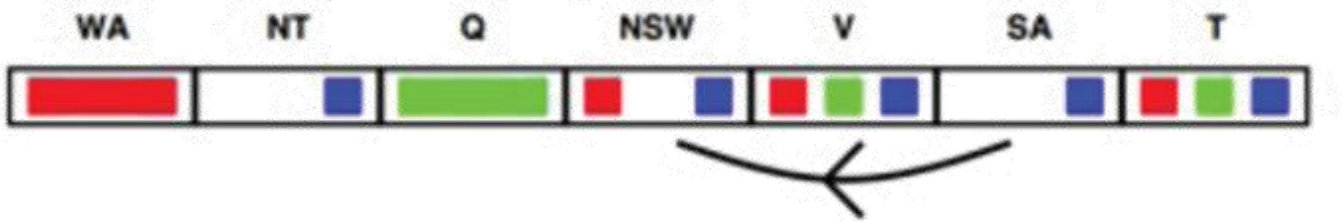
- **AC:** Simplest form of propagation makes each arc consistent
- $X \rightarrow Y$ is consistent **IFF** for every value x of X_i, there is some allowed y

Important: If *X* loses a value, neighbors of *X* need to be rechecked! Must rerun after each assignment!



SA → NSW

SA → NSW is consistent since *SA=blue* and *NSW=red* is a consistent assignment.

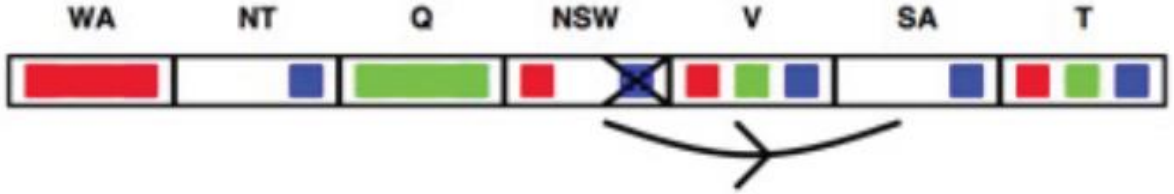


Arc Consistency

- **AC:** Simplest form of propagation makes each arc consistent
- $X \rightarrow Y$ is consistent IFF for every value x of X, there is some allowed y

NSW → SA

NSW → *SA* is not consistent since for *NSW=blue* there is no consistent assignment to *SA*.
Arc can be made consistent by removing *blue* from *NSW*



Arc Consistency

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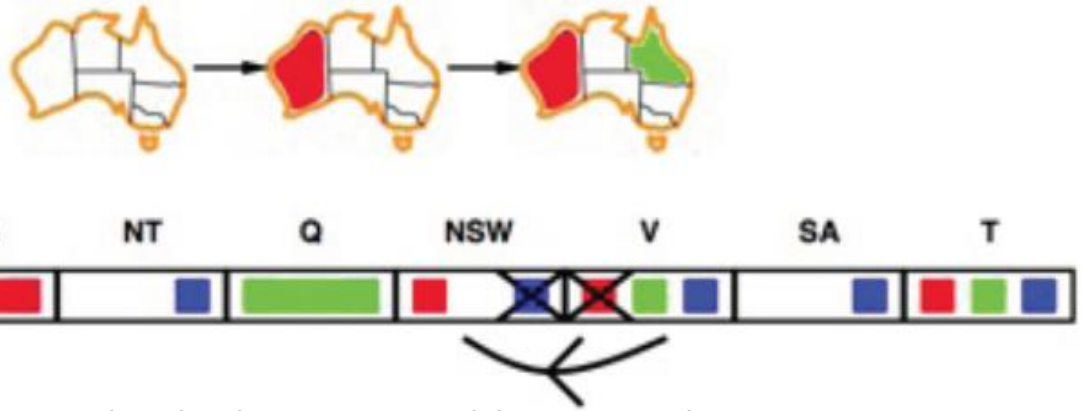
V → NSW



RECHECK neighbours !!

- Remove red from *V*

If X loses a value, neighbors of X need to be rechecked, some neighbors may become arc inconsistent

Arc Consistency

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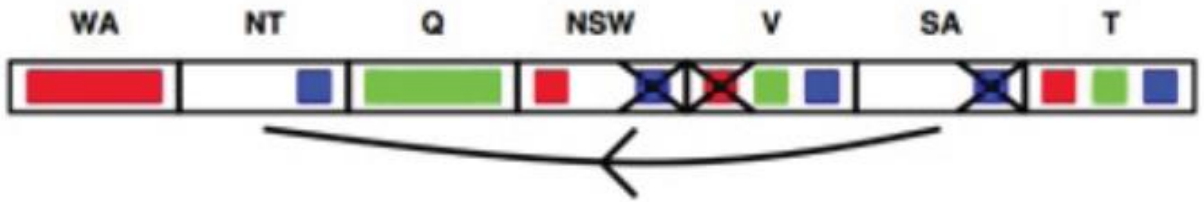
- $X \rightarrow Y$ is consistent IFF for everv value x of X. there is some allowed y

SA → NT

Can be run as a preprocessing before the search or after each assignment.

- Repeated until no inconsistency remains



Arc Consistency

```
function AC-3( csp) returns the CSP, possibly with reduced domains
   inputs: csp, a binary CSP with variables {X1, X2, ..., Xn}
   local variables: queue, a queue of arcs, initially all the arcs in csp

   while queue is not empty do
      (Xi, Xj) ← REMOVE-FIRST(queue)
      if REMOVE-INCONSISTENT-VALUES(Xi, Xj) then
         for each Xk in NEIGHBORS[Xi] do
            add (Xk, Xi) to queue
```

```
function REMOVE-INCONSISTENT-VALUES( Xi, Xj) returns true iff succeeds
   removed ← false
   for each x in DOMAIN[Xi] do
      if no value y in DOMAIN[Xj] allows (x,y) to satisfy the constraint Xi ↔ Xj
         then delete x from DOMAIN[Xi];  removed ← true
   return removed
```

Algorithm that makes a CSP arc-consistent!

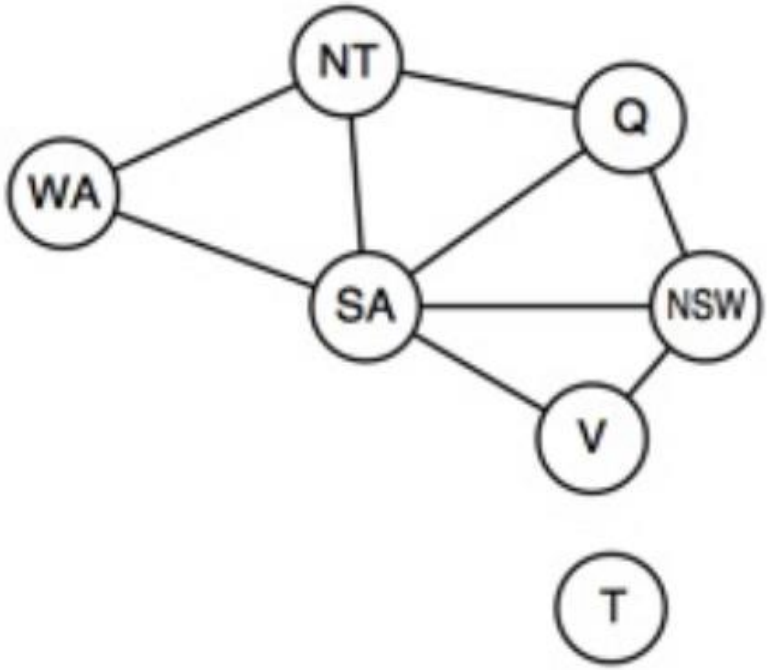
Time complexity: $O(n^2 d^3)$

Backtracking w/ inference

```
function BACKTRACKING_SEARCH(csp) returns a solution, or failure
    return BACKTRACK({}, csp)

function BACKTRACK(assignment, csp) returns a solution, or failure
    if assignment is complete then return assignment
    var = SELECT_UNASSIGNED-VARIABLE(csp)
    for each value in ORDER_DOMAIN_VALUES (var, assignment, csp)
        if value is consistent with assignment then
            add {var = value} to assignment
            inferences = INFERENCE(csp, var, value)
            if inferences ≠ failure then
                add inferences to assignment
                result = BACKTRACK(assignment, csp)
                if result ≠ failure then return result
        remove {var = value} and inferences from assignment
    return failure
```

Problem Structure



- **Idea:** Leverage the problem structure to make the search more efficient
- ***Example:*** Tasmania is an independent problem
- Identify the connected component of a graph constraint
- Work on independent sub-problems, which can reduce the complexity a lot

Problem Structure

Complexity

- Let $\boldsymbol{d}$ be the size of the domain and $\boldsymbol{n}$ be the number of variables
- Time complexity for BTS is $\boldsymbol{O}(\boldsymbol{d}^{\boldsymbol{n}})$
- Suppose we **<u>decompose</u> into sub-problems**, with c variables per sub-problem
  - Then we have $\frac{\boldsymbol{n}}{\boldsymbol{c}}$ sub-problems
  - c variables per sub-problem takes $\boldsymbol{O}(\boldsymbol{d}^{\boldsymbol{c}})$
  - The total for all sub-problems takes $\boldsymbol{O}\left(\frac{\boldsymbol{n}}{\boldsymbol{c}}\boldsymbol{d}^{\boldsymbol{c}}\right)$ in the worst case

Example: Problem Structure

- Assume n = 80, d = 2
- Assume we can decompose into 4 sub-problems with c = 20
- Assume processing at 10 million nodes per second

- Without decomposition of the problem we need:

$O(d^c)$ $2^{80} = 1.2 \times 10^{24}$ $nodes$ → when divided by 10 million = **3.83 million years!**

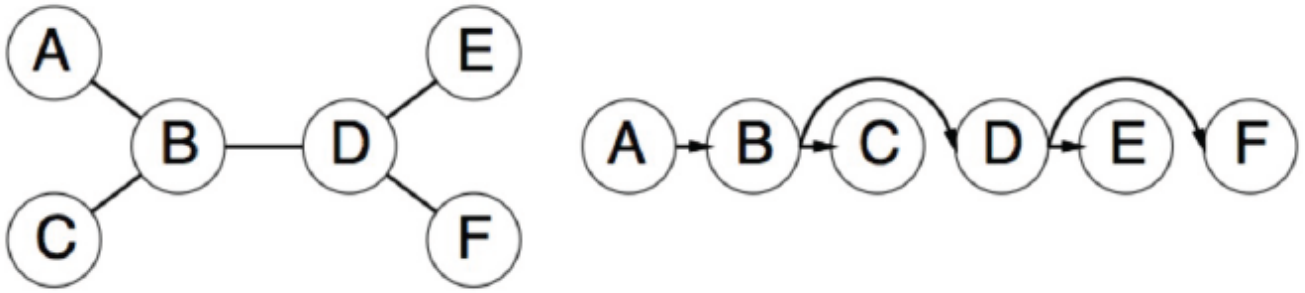
- With decomposition of the problem we need:

$O\left(\frac{n}{c} d^c\right)$ $4 \times 2^{20} = 4.2 \times 10^6$ $nodes$ → when divided by 10 million = **0.4 seconds!**

Problem Structure

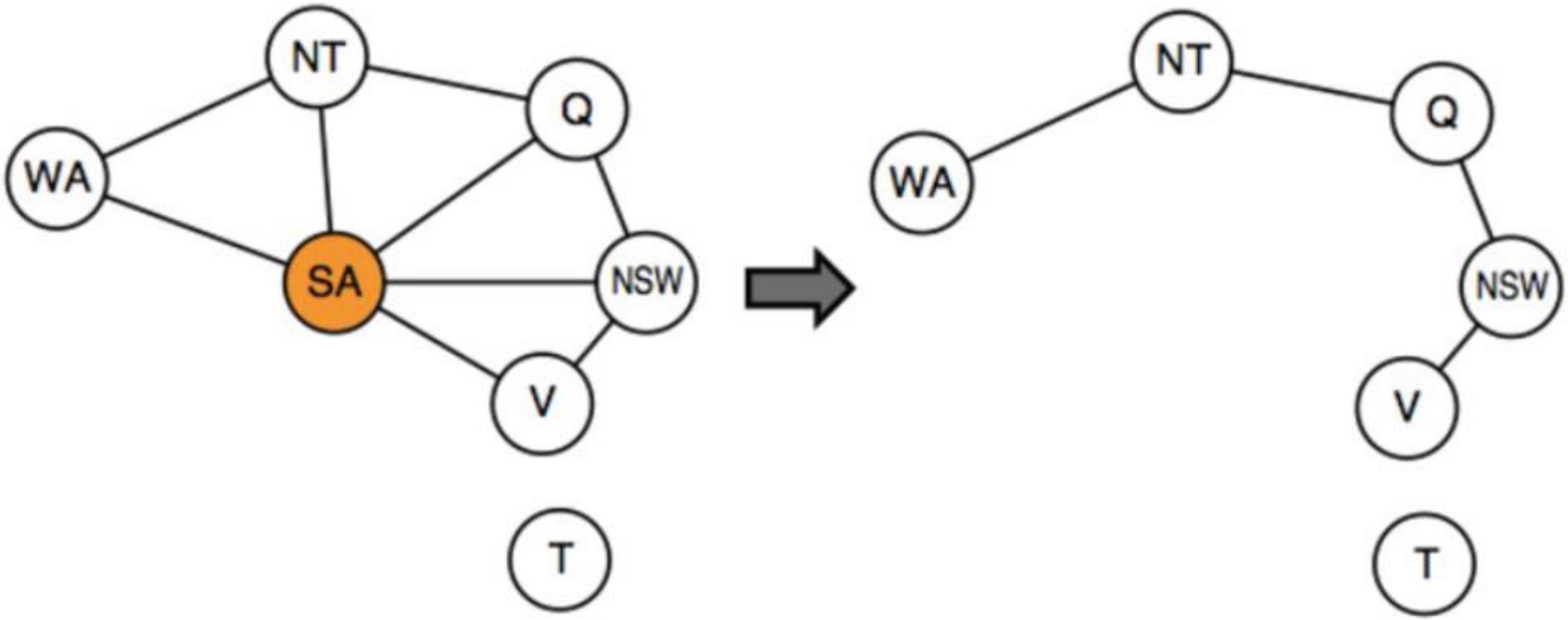
- Turning a problem into independent sub-problems is not always possible
- Can we leverage other graph structures?
  - Yes, if the graph is tree-structured or nearly tree-structured
  - A graph is a **tree** if any two variables are connected by **only one path**
- **Idea:** use DAC (Directed Arc Consistency)
- A CSP is said to be **directed arc-consistent** under an ordering $X_1, X_2, \ldots, X_n$ IFF every X_i is arc-consistent with each X_j for $j > i$

Problem Structure



- First pick a variable to the be the root
- Do a **topological sorting**: choose an ordering of the variables, *s.t.* each variable appears after its parent in the tree
- For n nodes, we have $n - 1$ edges
- Make the tree directed arc-consistent takes $\boldsymbol{O(n)}$
- Each consistency check takes up to $\boldsymbol{O(d^2)}$ (compare d possible values for 2 variables)
- The CSP can be solved in $\boldsymbol{O(nd^2)}$ - linear complexity

Nearly tree-structured CSPs



- Assign a variable or a set of variables and prune all the neighbors domains
- This will turn the constraint graph into a tree :)
- Assign {SA= x }
  - Remove any values from the other variables that are inconsistent
  - The selected value for SA could be wrong: have to try all of them

Summary

- CSPs are a special kind of search problems:
  - states defined by values of a fixed set of variables
  - goal test defined by constraints on variable values
  - assign values to the variables without violating the set of constraints

- Backtracking = depth-first search with one variable assigned per node
- Variable ordering and value selection heuristics help
- Forward checking prevents assignments that guarantee later failure

Summary

- Constraint propagation (e.g., arc consistency) is an important mechanism in CSPs
- It does additional work to constrain values and detect inconsistencies
- Tree-structured CSPs can be solved in linear time
- Further exploration: How can local search be used for CSPs?
- **The power of CSPs: domain-independent, that is you only need to define the problem and then use a solver that implements CSPs mechanisms**
- Play with CSP solver? Try ***http://aispace.org/constraint/***